

Detection 1 — Authentication Abuse

Detection Name:

Authentication Abuse / Brute Force Detection

Detection Goal:

Identify repeated failed authentication attempts against a Windows system to detect potential brute-force or password spraying activity based on threshold-driven analysis of Event ID 4625.

Log Source:

Windows Security Event Log

Event ID:

4625

SPL Query:

index=* EventCode=4625

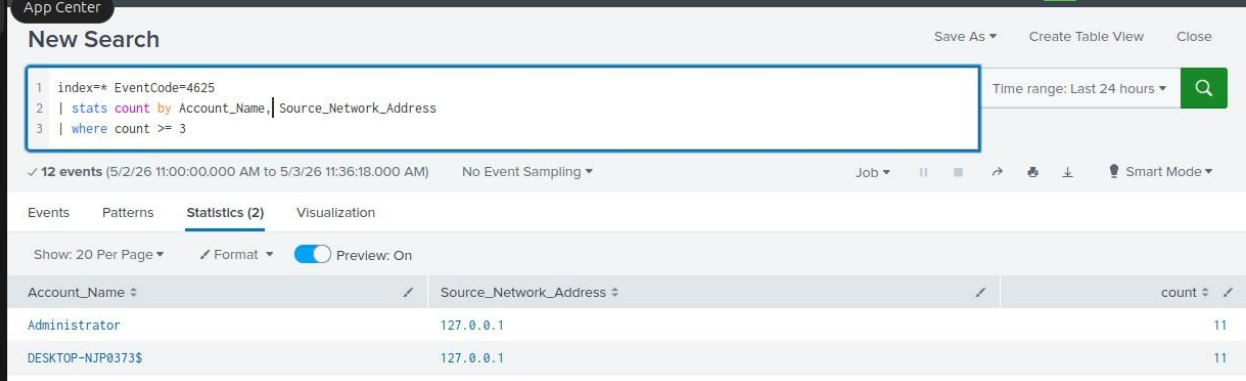
| stats count by Account_Name, Source_Network_Address

| where count >= 3

Test method:

I generated multiple failed login attempts on the Windows endpoint by intentionally entering an incorrect password several times using the same account. This simulated authentication abuse and allowed validation of Event ID 4625 in Splunk.

Results:



The screenshot shows the Splunk interface with a search query: `index=* EventCode=4625 | stats count by Account_Name, Source_Network_Address | where count >= 3`. The results are displayed in a table with 3 columns: Account_Name, Source_Network_Address, and count. Two rows are shown: Administrator with count 11, and DESKTOP-NJP0373\$ with count 11. Both are from the source network address 127.0.0.1.

Account_Name	Source_Network_Address	count
Administrator	127.0.0.1	11
DESKTOP-NJP0373\$	127.0.0.1	11

Interpretation:

The observed pattern of repeated failed login attempts against a single account within a short time frame is indicative of potential brute-force or password spraying activity. While this behavior can occasionally result from user error, the frequency and grouping of events suggest a

higher likelihood of unauthorized access attempts that warrant further investigation. **Next Steps:**

As an analyst, I would review subsequent successful login events (Event ID 4624) to determine if access was eventually gained. I would also check for account lockout events, identify whether the source of the login attempts is internal or external, and correlate this activity with other systems for additional context. If the behavior appears malicious or persistent, I would escalate the alert for further investigation and initiate appropriate containment actions.

Detection 2 — Suspicious Process Execution

Detection Name:

Suspicious Process Execution (Command Line / PowerShell) **Detection**

Goal:

Identify command-line and PowerShell activity on a Windows system that could indicate potential attacker behavior or unauthorized command execution.

Log Source:

Windows Security Event Log **Event**

ID:

4688 (Process Creation)

SPL Query:

```
index=* EventCode=4688
| search New_Process_Name="*cmd.exe*" OR New_Process_Name="*powershell.exe*"
| table _time host Account_Name New_Process_Name Process_Command_Line
| sort - _time
```

Test Method:

I generated process activity on the Windows endpoint by opening Command Prompt and PowerShell and running basic commands like whoami, ipconfig, and hostname to simulate normal user behavior and possible attacker actions.

Evidence:

New Search					Save As ▾	Create Table View	Close
<pre> 1 index=* EventCode=4688 2 search New_Process_Name="cmd.exe" OR New_Process_Name="powershell.exe" 3 table _time host Account_Name New_Process_Name Process_Command_Line 4 sort - _time </pre>					Time range: Last 24 hours ▾		
✓ 253 events (5/2/26 1:00:00.000 PM to 5/3/26 1:59:51.000 PM) No Event Sampling ▾					Job ▾		↻
Events Patterns Statistics (253) Visualization					Smart Mode ▾		
Show: 20 Per Page ▾ Format ▾ Preview: On					< Prev 1 2 3 4 5 6 7 8 ... Next >		
_time ▾	host ▾	Account_Name ▾	New_Process_Name ▾	Process_Command_Line ▾			
2026-05-03 13:19:46.369	DESKTOP- NJP0373	DESKTOP- NJP0373\$ -	C:\Program Files\SplunkUniversalForwarder\bin\splunk- powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk- powershell.exe"			
2026-05-03 13:19:45.547	DESKTOP- NJP0373	DESKTOP- NJP0373\$ -	C:\Program Files\SplunkUniversalForwarder\bin\splunk- powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk- powershell.exe" --ps2			
2026-05-03 13:19:42.628	DESKTOP- NJP0373	DESKTOP- NJP0373\$ -	C:\Windows\System32\dsregcmd.exe	"C:\Windows\System32\dsregcmd.exe" \$(Arg0) \$(Arg1) \$(Arg2)			
2026-05-03 13:01:14.762	DESKTOP- NJP0373	DESKTOP- NJP0373\$ -	C:\Program Files\SplunkUniversalForwarder\bin\splunk- powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk- powershell.exe" --ps2			

Figure 1: PowerShell and command-line process execution captured using Event ID 4688 in Splunk

Results:

The detection returned multiple process execution events showing both PowerShell and command-line activity, including command-line arguments associated with each process.

Interpretation:

Seeing PowerShell and command-line execution can be normal, but these tools are also commonly used by attackers for running commands, scripts, and post-exploitation actions. Because of this, repeated or unusual use of these processes could indicate suspicious activity.

Next Steps:

As an analyst, I would review the command-line arguments to understand what commands were executed and determine if the activity matches expected user behavior. I would also check for unusual patterns, correlate this with other logs, and investigate further if anything appears out of the ordinary or suspicious.

Detection Type:

Suspicious Process Execution (MITRE ATT&CK: T1059)

Detection 3:

Detection Name:

Network Reconnaissance via Command Execution

Detection Goal:

Identify the use of reconnaissance tools such as Nmap through process execution logs to detect

potential scanning or enumeration activity.

Log Source:

Windows Security Event Log

Event ID:

4688 (Process Creation)

SPL Query:

```
index=* EventCode=4688
| search New_Process_Name="*cmd.exe*" OR New_Process_Name="*powershell.exe*"
| table _time host Account_Name New_Process_Name Process_Command_Line
| sort - _time
```

Test Method:

I simulated reconnaissance activity from a Kali Linux system and generated command-line and PowerShell activity on the Windows endpoint to observe how process execution is logged.

Evidence:

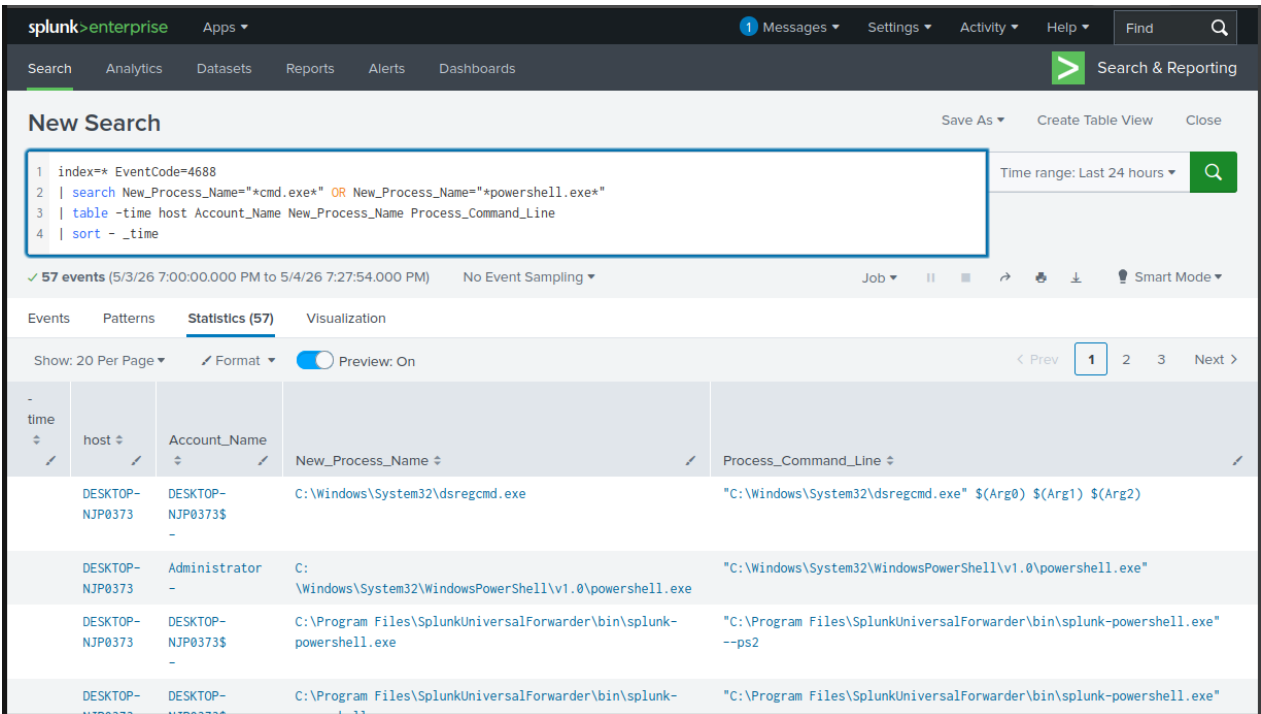


Figure 1: Command-line and PowerShell activity captured during simulated reconnaissance activity

Results:

The detection returned multiple process execution events involving command-line and PowerShell usage, including associated command-line arguments.

Interpretation:

Command-line and PowerShell tools are commonly used by attackers for reconnaissance, execution, and post-exploitation. Repeated or unusual use of these processes can indicate suspicious activity.

Next Steps:

As an analyst, I would review the command-line arguments, identify any unusual or unauthorized commands, and correlate this activity with other logs to determine whether it aligns with normal behavior or indicates potential malicious activity.

Detection Type:

Network Reconnaissance (MITRE ATT&CK: T1046)